Mechanical Engineering

	Normative 4-year Degree
Major	Mechanical Engineering 機械工程
Degree / Award Title	Bachelor of Engineering in Mechanical Engineering 工學士(機械工程)
Offering Academic Unit	Department of Mechanical Engineering
Normal Period of Study	4 years
Maximum Period of Study	8 years
Credit Units Required for Graduation	121 credit units

Note: The following curriculum information is subject to periodic review and changes.

GE Requirements (31 credit units) Catalogue Term : Semester A 2025/26

		Credit Units
GE University Requirements	GE1401 University English	3
	GE2410 English for Engineering	3
	GE1501 Chinese Civilisation - History and Philosophy	3
	GE1601 Whole-Person Development	1
GE Distributional Requirements	Take at least 1 course from each of the three distributional areas: Area 1: Arts and Humanities Area 2: Study of Societies, Social and Business Organisations Area 3: Science and Technology Visit the GE website for more course information.	12

College-specified Courses	MA1200 Calculus and Basic Linear Algebra I *Or* MA1300 Enhanced Calculus and Linear Algebra I	3
	MA1201 Calculus and Basic Linear Algebra II *Or* MA1301 Enhanced Calculus and Linear Algebra II	3
	MNE2066 Engineers in Society	3

College / School Requirements (6 credit units) Catalogue Term : Semester A 2025/26

Course Code	Course Title	Credit Units
CS1302	Introduction to Computer Programming	3
PHY1101	Introductory Classical Mechanics	3

Major Requirements (81 credit units) Catalogue Term : Semester A 2026/27

1. Core Courses (63 credit units)

Course Code	Course Title	Credit Units	Remarks
MA2001	Multi-variable Calculus and Linear Algebra	3	
MA3001	Differential Equations	3	
MNE2020	Engineering Workshop Practice	0	
MNE2029	Electrical and Electronic Principles I	3	
MNE2036	Engineering Computing	3	
MNE2109	Engineering Mechanics	3	
MNE2110	Engineering Materials	3	
MNE2112	Thermodynamics	3	
MNE2116	Engineering Graphics	0	# Department-specified Course for top 40% majors allocation
MNE3007	CAD/CAM	3	
MNE3010	Mechanical Design	3	
MNE3049	Control Principles	3	
MNE3058	Embedded Control Systems	3	
MNE3118	Mechanics of Materials	3	
MNE3119	Manufacturing Technology	3	
MNE3120	Measurement and Instrumentation	3	
MNE3121	Heat Transfer	3	
MNE3122	Fluid Mechanics	3	
MNE4010	Dynamics and Vibrations	3	
PHY1202	General Physics II	3	# Department-specified Course for top 40% majors

			allocation

Select **ONE** from the following:

Course Code	Course Title	Credit Units	Remarks
MNE4068	Project (Individual)	9	A course to fulfil the internship/ consultancy project/research project requirement
MNE4119	Project (Group)	9	A course to fulfil the internship/ consultancy project/research project requirement

2. Electives (18 credit units)

To fulfill the elective requirement, students can enroll in one of the following streams. They can take:

- **all 18 credit units from their chosen stream; OR**
- **a minimum of 12 credit units (2 "core" courses and 2 electives) from their chosen stream, and the remaining credit units from the common elective pool.**

There is no GPA requirement for enrolling in a stream.

Robotics and Mechatronics Stream Requirements

Course Code	Course Title	Credit Units
MNE3046	Automation Technology	3
MNE4032	Robotics and Machine Vision	3

Choose at least two of the following courses:

Course Code	Course Title	Credit Units
CS2311	Computer Programming	3
MNE2204	Aircraft System Design	3
MNE3123	Internship in Engineering	3
MNE4121	Machine Learning and Quantum Computation	3

Materials and Mechanics Stream Requirements

Course Code	Course Title	Credit Units
MNE3063	Material Analysis for Product Quality	3
MNE3127	Electron Microscopy	3

Choose at least two of the following courses:

Course Code	Course Title	Credit Units
CS2311	Computer Programming	3
MNE3123	Internship in Engineering	3

MNE4005	Finite Element Analysis	3
MNE4121	Machine Learning and Quantum Computation	3

Thermofluid and Energy Stream Requirements

Course Code	Course Title	Credit Units
MNE3104	Sustainable and Renewable Energy	3
MNE3129	Experimental Sustainable Engineering Techniques	3

Choose at least two of the following courses:

Course Code	Course Title	Credit Units
CS2311	Computer Programming	3
MNE3123	Internship in Engineering	3
MNE3125	Fluid Mechanics for Energy-Related Applications	3
MNE4130	Introduction to Multiphase Flow Science	3

For students who do not want to focus on a selected stream, they can take:

- **all 18 credit units from the following list; OR**
- **a minimum of 12 credit units from the following list, and the remaining credit units from the common electives pool.**

Course Code	Course Title	Credit Units
CS2311	Computer Programming	3
MNE2204	Aircraft System Design	3
MNE3046	Automation Technology	3
MNE3063	Material Analysis for Product Quality	3
MNE3104	Sustainable and Renewable Energy	3
MNE3123	Internship in Engineering	3
MNE3125	Fluid Mechanics for Energy-Related Applications	3
MNE3127	Electron Microscopy	3
MNE3129	Experimental Sustainable Engineering Techniques	3
MNE4005	Finite Element Analysis	3
MNE4032	Robotics and Machine Vision	3
MNE4121	Machine Learning and Quantum Computation	3
MNE4130	Introduction to Multiphase Flow Science	3

Common Electives

Course Code	Course Title	Credit Units	Remarks
FS2001	Workshop-based Study in Science and Engineering	3	
MNE3126	Methodologies for Solving Complex Engineering	3	

	Problems		
MNE3128	Radiological Analysis	3	
MNE4047	Directed Studies	3	
MNE4109	Reliability Engineering and Risk Analysis	3	
MNE4120	Bioinspired Composites: Design, Mechanics and Manufacturing	3	
MNE4125	Principles and Transport Process Fundamentals of Semiconductor Manufacturing	3	
MNE5112	Mechanical Design with Advanced Material & Additive Manufacturing	3	Only for final year undergraduate students (with quota restrictions)
SYE3102	Quality Engineering	3	
SYE4024	Project Management	3	

3. Optional Courses

Course Code	Course Title	Credit Units	Remarks
FS2002	Industrial Attachment Scheme	3	Internship (minimum 6 weeks)
FS3002	Industrial Attachment Scheme	3	Internship (minimum 6 weeks)
FS4001	Co-operative Education Scheme (CES)	8	Internship (8 - 12 months)
FS4002	Industrial Attachment Scheme	3	Internship (minimum 6 weeks)
FS4005	Overseas Internship Scheme	3	Internship (9 - 13 weeks)